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Questioning trust in AI research: exploring the influence of trust assessment on dependence in AI-assisted decision-making

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ABSTRACT
Trust is considered crucial for effective interaction between humans and artificial intelligence (AI), necessitating valid trust assessment methods. The 'question-behaviour effect', however, suggests that by applying a questionnaire subsequent behaviour can be influenced, for example participants' dependence on AI. The objective of the present research was to examine the effect of trust assessment on reliance in the context of an AI-supported decision-making task. We designed an AI-supported task, requiring participants to decide on patterns in so-called Kandinsky Figures. In a scripted experiment with a 2×2 between-subjects design, $N = 149$ participants' trust was assessed at different times (before block 1 or block 2) and with different assessment extent (i.e. scale length). Participants' agreement with AI recommendations and task completion time served as behavioural trust indicators. We found no effect of trust assessment on behaviour and correlations between trust and dependence were notably low. Participants' dependence matched the instructed reliability level of the AI system and our findings did not suggest the presence of a question-behaviour effect of trust assessment. Overall, while the conduction of trust assessment did not influence dependence, our results question the conceptualisation of trust as a general predictor for dependence, especially in comparison to instructed reliability.

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KEYWORDS
Human-AI interaction; explainable artificial intelligence; trust in AI; trust assessment

1. Introduction

Systems based on artificial intelligence (AI) affect the world we live in, as they analyze data, suggest decisions, or make predictions (Duan, Edwards, and Dwivedi 2019). Thereby, AI systems can influence an individual's life crucially, for example, in medical therapy contexts (Waring, Lindvall, and Umeton 2020), by predicting liability for loans (Arun, Ishan, and Sanmeet 2016) or serving as the basis for governmental decisions (Hadwick and Lan 2021). Accordingly, calls for reliable and safe AI have been numerous (e.g. Shneiderman 2020) and the first policies for AI regulation have been positioned (see Veale and Zuiderveen Borgesius 2021). However, determining the actual reliability of an AI system can be a challenging task for institutions and individuals, due to the opacity of systems as well as the lack of skills or resources on the part of the user (Rieger, Roesler, and Manzey 2022). In such situations, research has shown that individuals approach the system by assuming certain reliability (J. L. Wright, Chen, and Lakhmani 2019). If this assumption is incorrect, errors in AI systems could lead to increased compliance by the individual (Madhavan and Wiegmann 2007). Hence, the interaction of AI and humans leads to challenges for human-system trust, well known from human-automation research: without knowing how good the AI may perform, users are vulnerable when their expectations are violated. In other words, given the uncertainty and vulnerability in interactions with AI trust might be a central component shaping how individuals behave (see Hoff and Bashir 2015). Research in human-AI interaction (HAII) focuses on understanding how AI system design can enable individuals to easily assess the extent to which they can trust a system. To understand human trust in AI, trust measurement needs to be reliable and validated.

In HAII research, the conceptualisation of trust towards AI systems often refers to existing work on trust in automation (Chiou and Lee 2023; Hoff and Bashir 2015; Muir 1994), as AI represents automated information processing with varying levels of task allocation between AI and human users (Paul et al. 2022).

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Trust is commonly defined as an attitude, and a direct assessment thus requires a subjective rating, such as in the form of self-reported trust (Kohn et al. 2021). Oppositely, research also aims to use indicators such as behavioural (D. Miller et al. 2016) or physiological (Hergeth et al. 2016) signals, for example, because they can be sampled more frequently and are less intrusive. Moreover, previous studies have repeatedly shown, that the relation between self-reported trust, reliability of systems, or dependence can be low (Elder et al. 2022; Wiczorek and Manzey 2010). Hence, further research on the conditions that influence the relationship between self-reported trust and behaviour, is needed (c.f. Bustamante 2009; Patton and Wickens 2024).

In addition to the obvious lack of predictive power, another disadvantage of self-assessments in trust research is that they can be intrusive and alter the participants' response in an experiment. The observation that a questionnaire can change people's behaviour is described as the question-behaviour effect (QBE). Several studies have shown that measuring an attitude can influence behaviour (Bone et al. 2017; Chandon, Morwitz, and Reinartz 2004; Zwane et al. 2011). In particular, the measurement of trust could trigger introspection that affects people's behaviour (Wilson and Schooler 1991). Accordingly, the exact timing of the trust measurement during the experiment could be a key factor in explaining the different levels of correlation between self-reported trust assessment and trust-related behaviour (see van Dongen and van Maanen 2013). Not only the occurrence of a measurement, but also its scope (i.e. the number of items) could influence subsequent behaviour. Repeated reflection on multiple items compared to, for example, a single item could influence a person's attitude, comparable to the effect of mere repeated exposure (Zajonc 1968). As far as we are aware, there are no studies on the measurement effects of self-reported trust assessment on trust related behaviour. However, such results are needed to ensure internal validity and to clarify existing questions in the interpretation of experimental studies (see Bolton 2022). Since trust is not subsequently assessed in real-world contexts, it would be problematic if experiments systematically lost their ecological validity as a result.

Accordingly, the objective of the present research was to examine the QBE of self-reported trust assessment on dependence in the context of an AI-supported task. To this end, we designed an abstract pattern recognition task based on current paradigms in AI research (Muller and Holzinger 2021) and varied at what point in time and to what extent trust was surveyed from the participants. We aimed to investigate the presence of a question behaviour effect of trust on dependence, and explore how characteristics such as the timing and length of the measurement may impact this effect.

2. Background

2.1. Trust as an attitude

Whereas trust is commonly described via different concepts such as beliefs, attitudes intentions, or behaviour in the literature (Chiou and Lee 2023), there is the overarching consensus in HAII that trust should be conceptualised as an attitude rather than a behaviour (especially based on Lee and See 2004). As an attitude, trust is an inner tendency of how to evaluate an entity, and therefore trust is not directly observable. This is closely mirrored in the definition from Lee and See (2004, 51), defining trust as an 'attitude that an agent will help achieve an individual's goals in a situation characterised by uncertainty and vulnerability'. Also, as reflected in multiple models of trust (Chiou and Lee 2023; Hoff and Bashir 2015; Lee and See 2004), existing HAII research treats trust as a multidimensional concept. However, in various studies in HAII research trust is currently assessed as an unidimensional attitude that can be utilised to predict user behaviour such as and especially dependence (Bergkvist and Rossiter 2007; De Vries, Midden, and Bouwhuis 2003). The relevance of trust as a construct for HAII (Bolton 2022; Choung, David, and Ross 2023) and the fact that trust as an attitude can only be assessed through selfreported assessments is currently discussed (Brzowski and Nathan-Roberts 2019).

Self-report methods of assessing trust are the most common way of assessing trust in research on HAII (Kohn et al. 2021). Self-report measures of trust mainly consist of self-rated scales, with Jian et al.'s 'Trust in Automation' scale being the most cited (Jian, Bisantz, and Drury 2000). However, while there are several models of trust in automation (see e.g. Lee and See 2004; Mayer and Davis 1999) and scales based on these models (e.g. Korber, Baseler, and Bengler 2018), most are self-constructed and custom scales that are usually not explicitly validated or grounded on theoretical models (Kohn et al. 2021; Razin and Feigh 2024). In addition, applied research often aims to achieve economic assessment methods and therefore, relies on single-item scales. However, to improve the methodological reliability of self-reported trust assessment, several studies assess trust with longer scales, for example, (Jian, Bisantz, and Drury 2000; Korber, Baseler, and Bengler 2018). Furthermore, existing self-report measures of trust address heterogeneous constructs:

they position trust as an attitude toward a trustee (Jian, Bisantz, and Drury 2000; Korber, Baseler, and Bengler 2018), as a perception of a system property such as trustworthiness (Trommler, Attig, and Franke 2018) or reliability (Merritt et al. 2015). Due to the complexity and lack of standardisation in the assessment of trust, as well as the high number of individual assessment methods in the literature, the labelling of trust scales as 'measures' of trust could be criticised, as the term measure in its strict meaning is thought to refer to standardised methods that define the extent of a property and are sufficiently validated (Razin and Feigh 2024; B. D. Wright 1999). That is, while many research papers use the term 'trust measure' we refer to the term 'trust assessment' in the present paper. Additionally, note that we will use the term 'dependence' for behaviour related to following system recommendations in line with (Meyer 2004).

Due to the challenges of their psychometric quality, the substitution of self-reported trust assessment with behavioural methods could support the standardisation and comparability of trust research. However, while existing models link, for example, dependence to trust (Lee and See 2004), dependence can be seen as a possible behavioural consequence of trust (comparable to, e.g. indirect or implicit methods to assess attitudes, see Cunningham, Preacher, and Banaji 2001). Dependence can be analyzed by indicators of consistency, performance, behaviour (such as verification), or response bias (Wang, Jamieson, and Hollands 2008), which compose methods where the actions of a participant are observed.

In addition, while existing research approaches the assessment of trust through physiological methods, it remains unclear what psychological variables these measures truly capture (Kohn et al. 2021). Other influences such as risk or workload might affect physiological reactions and behaviour, thus, overshadowing effects of participants' trust. Moreover, physiological measures can be expensive (e.g. Hergeth et al. 2016 utilising eye-tracking) and may not be suitable for field research. Accordingly, self-report measures of trust are likely to remain the primary method for assessing trust. However, research is needed on the relationship between different trust assessment methods and their implications for experimental design.

2.2. The relationship between trust & dependence

The relationship between trust and dependence as a behaviour influenced by trust, is discussed among multiple technologies and domains (Dzindolet et al. 2003; Kerasidou et al. 2022; Solberg et al. 2022). Exemplary, studies cited by Lee and See (2004) on trust and dependence are based on correlation analysis (see, e.g. Muir and Moray 1996; Patton and Wickens 2024). For example, when the self-reported trust assessment takes place after participants could choose to rely on a system or not rely on it, the self-reported trust assessment could only reflect how they perceived their behaviour. That is, self-reported trust assessment would be highly correlated with dependence, but could not be used as a predictor (see Bustamante 2009; Wiczorek and Manzey 2010). While several empirical studies on trust and dependence assume a predictive value of trust assessment for behaviour (De Vries, Midden, and Bouwhuis 2003; Merritt et al. 2013; Merritt et al. 2014), other studies of dependence in automation demonstrate results that indicate a more complex connection, questioning the predictive value of trust assessments (Hoesterey and Onnasch 2023; Papenmeier et al. 2022; Patton and Wickens 2024; Ueno et al. 2023). The reason for that may be additional factors that influence behaviour but potentially do not affect trust in the same way, for example, workload or the extent of current risk. In addition, the quantity of studies that combine both, self-reported trust assessments as well as behavioural methods to quantify dependence is scarce, especially compared to studies that rely only on self-reported trust assessment (Kohn et al. 2021). However, studies integrating both, dependence and self-reported assessments, usually try not to understand the predictive value of trust for dependence. In order to do so, those studies are forced to assess self-reported trust before dependence is measured. The effect of a self-reported trust assessment on dependence, however, has not been studied sufficiently to guarantee the internal validity of the results found in those studies.

In previous studies, changes in dependence as a reaction to AI errors were dependent on the type of error the system made (Sanchez et al. 2011; van Dongen and van Maanen 2013). Accordingly, the predictive value of trust assessment for dependence could be affected by the experimental sequence. (van Dongen and van Maanen 2013) modified, whether automated aid in a decision task was shown before or after participants made their own decision. They demonstrated that assessed trust may be more influential on dependence when participants have to make an initial decision, but not when the decision aid is presented first. In any case, the sequence of assessing trust and measuring dependence may affect study results. The experimental conditions under which trust assessment and dependence interact have not been sufficiently studied yet. Hence, sequential assessment effects could be the reason for contradictory or at least inconsistent results.

2.3. Assessing attitudes may affect behaviour

Potential effects of assessing trust on trust-related behaviour can be explained by different theoretical approaches: On the one hand, previous research, e.g. involving the mere-measurement effect (Morwitz and Fitzsimons 2004), suggests that measuring an attitude can increase behaviour consistent with that attitude (Godin et al. 2008). It is suggested that this effect is due to the fact that reflecting on the attitude makes it more cognitively accessible, thereby increasing the likelihood of related behaviour (e.g. purchase decisions, Fitzsimons and Williams 2000). Consequently, asking a person about their trust in a system might make them more likely to engage in behaviour that demonstrate trust.

While it is possible that the mere-measurement effect also occurs in trustrelated behaviour, its measurement in the context of HAII is difficult to demonstrate: trust as an attitude is characterised by more dynamics than, for example, the attitude towards blood donation or purchasing decisions, which were associated with the mere-measurement effect in earlier research. In empirical research on trust in AI, the attitude towards certain systems or robots is investigated, which is, however, interaction-dependent and less stable. As a result, a temporally separate measurement is rarely found in research on trust in AI. While in experiments the asessment of trust can also be recorded at the beginning or end, in the mere measurement study the attitude is sometimes measured weeks in advance. In other words, while attitudes towards fundamental decisions such as organ donations are relatively stable, thus, less dependent on the current situation (e.g. a task or a performance that has just been shown) than the construct of trust.

In the literature on the mere measurement effect (which unfortunately uses other terms such as QBE synonymously in some cases, see Sarah Wilding, Conner, and Sheeran 2016), explanatory approaches are given that can also be applied to trust. (Song and Schwarz 2008) describe that thinking about behaviour may activate cognitive mechanisms such as processing fluency and behavioural simulation. That is, the mental representation of a behaviour in which one controls a system or manages the precision of a system may make one more likely to engage in that behaviour. For example, behaviour associated with trusting or controlling a system may be cognitively activated and thus more likely to be performed, regardless of whether they are consciously intended. Previous research suggests that this cognitive activation increases related behaviour in general by making it more accessible. The latter described effects are less specific than the mere measurement effect, as trust-related behaviour is not dependent on the individuals' attitude (and its valence) but is generally more accessible. However, the predictions made by the question-behaviour effect are less precise than the mere-measurement effect, which focuses specifically on the alignment of attitudes and behaviours (Morwitz and Fitzsimons 2004). Consequently, the QBE might be more suitable for a more dynamic construct like trust.

Based on previous literature on experimental effects of attitude assessment, we suggest that the introspection, triggered by the evaluation process (i.e. a questionnaire), may slow response times and reduce confidence in decisions, as suggested by (Wilson and Schooler 1991) and aim to investigate it in an experimental study. Here, it is crucial to investigate whether trust assessments are the primary driver of behavioural change or whether other factors, such as perceived system reliability, play a more important role.

3. Present research

The objective of the present research was to evaluate the presence of an experimental artifact in human-AI trust research, i.e. effect of trust assessment on dependence in the context of human-AI interaction. Based on previous findings in the literature, we argue that a general question-behaviour effect could systemically affect user behaviour in experimental studies.

Our first research question focuses on the presence of the question-behaviour effect. In particular, we expected higher dependence for the control group (no trust assessment) compared to the experimental groups at both early and late assessment (H1). Moreover, we aimed to investigate if the extent of assessment influences dependence. We expected lower dependence after a multi-item assessment compared to a single-item assessment (H2). Lastly, we were interested in the influence of time of assessment on dependence. We assumed that participants demonstrated lower dependence overall when trust was assessed at an early point in time of the experiment compared to an assessment at a later time (H3).

In addition, to understand what other variables may influence human dependence, we included a regression analysis of perceived reliability, instructed reliability, and trust assessment. As experience with the system could also affect confidence, we examined the development of confidence over time, independent of the experimental condition. The present research was preregistered (removed for anonymisation).

During the review process, we realised that we could also try to test the mere measurement effect described

above, at least to some extent, on the basis of our data. Therefore, we conducted an exploratory investigation to see whether we could better interpret the behavioural effects found if we included the level of the attitude (trust). To do this, we had to assume that trust in the groups without trust assessment is distributed similarly to the group with trust assessment.

Ethics approval for this study was granted by the Ethics Committee of the Local University before the start of the experiment.

4. Method

4.1. Participants

149 participants were recruited via the online platform prolific (Prolific Team, Prolific, Prolific Academic Ltd, London, United Kingdom 2014). Based on our preregistered exclusion criteria (i.e. completion time shorter or longer than median participation time ±2 standard deviation, lack of variability in responses, and correct responses below 20% in task block 1 and task block 2) no participant needed to be excluded. Participants were between 19 79 years (*Mean* = 43.9, *SD* = 14.4). 98 participants identified themselves as female, 59 as male, and one participant preferred not to answer. All participants were residents of the United Kingdom. Additionally, the affinity for technology interaction (ATI) of the participants was assessed ($M = 3.24$, $SD = 1.04$). The ATI examines the individual tendency to interact with technology, which is why a distribution that deviates from larger samples could be the basis for effects. This is not to be assumed in our sample (Franke, Attig, and Wessel 2019).

Participants volunteered to participate in the study, and informed consent was obtained. Only native English speakers were recruited for the study. The study was conducted with LimeSurvey (LimeSurvey Project Team / Carsten Schmitz 2012). Participants were instructed to conduct the study only with appropriate screen sizes, i.e. on desktop computers, laptops, or tablets. Participants were compensated £3.50 for their time in the study. In addition, every participant with a performance over 90% was awarded an additional £2. This performance-related reward was offered to incentivise and motivate participants beyond their general compensation.

4.2. Task & material

In the experiment, we utilised Kandinsky Patterns, which are sets comprised of individual images called Kandinsky Figures (KF). Kandinsky Figures were introduced by (Muller and Holzinger 2021) and describe images consisting of squares containing 1 to *n* geometric objects. A set of Kandinsky Figures that follows the same rule (e.g. there are more red than green objects) can be described as a Kandinsky Pattern. Following (Muller and Holzinger 2021) our implementation of Kandinsky Pattern can be described as follows: each Kandinsky Figure is a square image that contains 1 to *n* objects defined by their sizes, colours, shapes, and positions in this image. A statement about a KF can be denoted in natural language, linguistically, or mathematically and can either be true or false. A Kandinsky Pattern is defined as a subset of all KF for which a certain set of statements, which we call rules, apply.

We developed a unity-based application that allows for the systematic creation of images of KF (KFgen app). The app can be found and used under https://github.com/JonasJakobi/Kandinsky to create stimuli like the ones used in the present experiment. In the KFgen app, the experimenter can specify which rules should apply to all generated KF. In the present research and version of the KFgen app three possible types of rules can be created: Rules relating to the relative positioning of objects, for example, 'Red objects above green objects', rules restricting the maximum amount of a certain type of object, for example, 'There can only be two squares', and rules that force a minimum amount of an object, for example, 'There have to be more than two circles'. KFgen produces a set of KF, randomising each KF while maintaining the rules. For the rules to stay deducible, each KF will contain at least one of each relevant object (i.e. all objects referred to in the rules). So with a rule like 'Squares above triangles', the generated KFs will always contain at least one square and triangle.

In the KFgen app rules can also be inverted. This allows for the generation of a KF that does not follow the same rules as the other KFs and therefore does not belong to the corresponding Kandinsky Pattern. Instead of just not following the rules, these unfitting KFs will instead explicitly break the rules by also containing each relevant object to break a rule.

We used Kandinsky Patterns as the experimental context for our research. Kandinsky Tasks (KT) are based on the described KFs and were designed as follows: a total of five KFs were displayed. Four of them followed a Kandinsky Pattern, i.e. an underlying rule. A fifth picture explicitly violated this rule. For example, four out of five depicted KFs only contained red objects above green objects. A fifth picture, in contrast, contained red objects under green objects and was therefore violating the rule 'All objects are above green objects'. These five pictures were displayed horizontally next to each other in a random order. The pictures were labelled

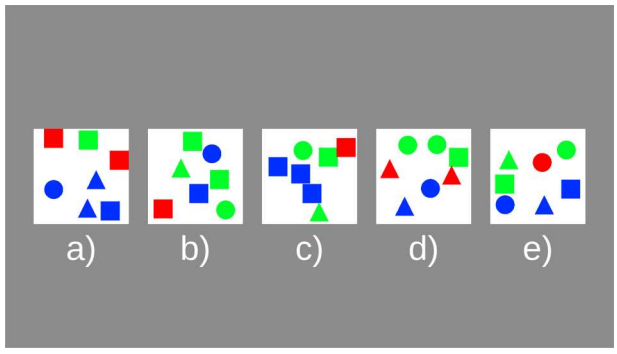

Figure 1. Depiction of a set of five Kandinsky figures. Four of them follow the rule that every red object has to be above blue objects (a Kandinsky Pattern). However, the second Kandinsky Figure, labelled (b) does not follow that pattern. Images like these were used to assess participants' baseline performance.

(a)–(e) as shown in Figure 1. The participants' task in the KT was to select the picture that violated the rule. In our experiment, rules were not change for 10 subsequent tasks, i.e. the same rule applied for multiple tasks after each other (similar to the game Mastermind, Stuckman and Zhang 2005). Participants did not have the opportunity to revisit previous rules or decisions. Whenever a rule was changed, the previous rule was revealed to participants.

For the trials with AI support, an additional text cue was generated. This read 'The AI recommends x', where x was the letter of the picture that violated this rule. In all groups, the AI's reliability was 100% throughout all experimental trials, i.e. the textual cue was always correct.

KT provide a great basis to research HAII, as they canpotentially be AI-assisted, as well as solved by humans. Furthermore, KT are sufficiently abstract that participants cannot benefit from pre-existing knowledge, compared to, for example, studies in the medical domain. To understand the effects of human dependence, the task can be performed correctly, i.e. there is an objectively correct answer, that allows the experimenter to determine performance.

In addition, stimuli are also well suited for investigating the effects of trust assessment on subsequent behaviour: although the task is challenging, participants can check for themselves whether they want to comply with the system's suggestions, as they have access to all the information relevant to the task. In order to do this, however, considerably more time must be used in some cases, as all images must be checked in one go. Hence, there is therefore an interest on the part of the participants in complying with the AI and saving resources, but this is not absolutely necessary. In addition, it can always be clearly determined whether the people have followed the system's suggestion or not.

4.3. Pre-study

We conducted a pre-study with $N = 30$ participants recruited via prolific. Participants were between 22 - 67 years ($M = 37.7$, $SD = 11.3$). 17 identified themselves as female, and 13 as male. All participants were residents of the United Kingdom ($n = 29$) or the United States ($n = 1$). In particular, the task's difficulty was tested. Within the pre-study, the average percentage of correctly answered Kandinsky Tasks of the participants without AI support was 62% and with AI support close to 90%. Within the pre-study, the AI's reliability was stated as 90%, whereby it was 100% correct. As the study in the pre-study lasted over 1.5 h in some cases, the second block was shortened by 10 stimuli in the main study. The discrepancy between instructed reliability and the system's true reliability was increased, to support users to recognise the difference despite the rather short overall period of experience with the AI system. Accordingly, the system was introduced with only 80% reliability in the main study. In addition, based on an effect size of $\eta^2 = 0.038$ found between task completion time between groups in the first task block and aiming for 80% power, we calculated a required sample size of 150 participants for the main study (based on G*power, Faul et al. 2009).

4.4. Study design

The study followed a 2 × 2 design, with the time of trust assessment (early vs. late; i.e. trust assessment prior to block 1 or prior to block 2) and extent of trust assessment (i.e. single-item or multi-item scale) as manipulated variables, and a control group as depicted in Figure 2. Accordingly, each participant was assigned to one of five conditions:Control condition (Con), early trust assessment, single item (E-S), early trust assessment, multi-item (E-M), late trust assessment, single item (L-S), or late trust assessment, multi-item (L-M).

4.5. Dependent variables

Trust. In the multi-item conditions, trust was assessed using the Trust in Automation (TiA) scale from Jian, Bisantz, and Drury (2000) as it is one of the most frequently used scales for self-reported trust assessment (Kohn et al. 2021). Accordingly, for the single item condition, we used the following item: 'How much do you trust the system?' rated on a six-point Likert scale from 1 (not at all) to 6 (completely), to compare it to

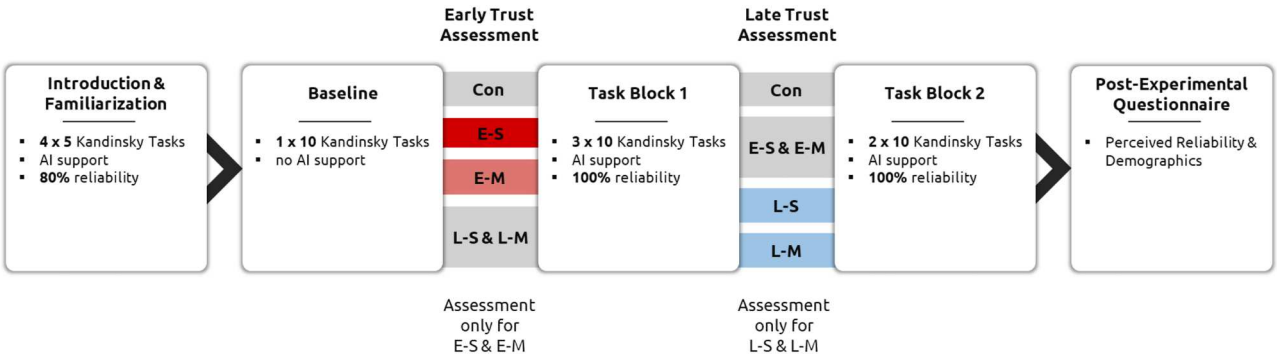

Figure 2. Depiction of study design, highlighting the different points in time for trust assessment.

the TiA scale as done in previous research (e.g. Rieger, Roesler, and Manzey 2022). We decided to use a six-point Likert scale in both cases instead of a scale ranging from 0 to 100 (single item as used in Rieger, Roesler, and Manzey 2022) or 1 to 7 (multi item as originally introduced by Jian, Bisantz, and Drury 2000) compared to previous research, to achieve higher consistency between the response scales used in our study.

Dependence. We measured dependence based on two variables (1) agreement between users and AI and (2) task completion time. First, the extent to which people followed the AI recommendation for their answers was calculated. One commonly used measure to quantify the agreement of two raters is Cohen's Kappa coefficient (Blackman and Koval 2000). Cohen's Kappa compares the observed agreement between the AI recommendation and user responses with the expected agreement that would occur by chance alone. It takes into account both the proportion of agreement and the possibility of agreement by random chance. The possible values of Cohen's Kappa range from −1 to 1. A positive value indicates agreement beyond chance, where 1 indicates perfect agreement, and 0 indicates agreement equal to chance. For consistency, we report Cohen's Kappa in percent. Second, task completion time was computed by capturing how long each participant spent in a block divided by the number of presented stimuli in each block (i.e. the average time for one Kandinsky task, i.e. five pictures to compare, in each block).

4.6. Procedure

After obtaining informed consent, KF and tasks were explained. Participants were then asked to familiarise themselves with the task. They were presented with 20 Kandinsky tasks, with the Kandinsky pattern changing after every fifth task. The rule was shown to the users and they only had to decide whether the AI recommendation shown was correct or not. The AI system's reliability was introduced to the participants at 80%. In order to have a perfect fit between description and experience the AI's reliability in this introductory section was also scripted to be 80% (Haines, Kvam, and Turner 2023). Afterwards, during the Baseline block the participant's ability to correctly identify the unfitting image without AI support was assessed. They were presented with ten Kandinsky tasks following the same rule (red objects must be below blue objects). The rule was revealed to participants after they had completed all ten Kandinsky tasks. After the baseline phase before the actual experiment started, a trust assessment was conducted for both groups in the early condition.

In the following first experimental block, participants were presented with 30 Kandinsky tasks. The rules on which the KFs were based changed after ten tasks, i.e. three different rules were presented in this block. The order of the rules presented was randomised within the block. Participants received feedback on each rule after completing ten tasks, thus, they were able to understand whether the AI made correct recommendations or not. Next, a trust assessment was administered to all groups assigned to the late trust assessment conditions. The same procedure as in block 1 was repeated, but only with 20 Kandinsky tasks and two different rules. Again, the order of the rules presented was randomised and the rules were revealed after the tasks had been completed. After completing block 1 and block 2, participants were asked to estimate the AI's reliability in percent as an indicator of perceived reliability. Variables to assess interpersonal differences were also administered at this point. Finally, participants were given a code to receive compensation.

5. Results

5.1. Descriptive & baseline values

We calculated the baseline values of participants' ability in the Kandinsky task that were not supported by AI for dependence and task completion time. We then

Table 1. Descriptive statistics of dependence assessed through Cohen's kappa coefficient, time on task measured in seconds, and trust for each condition and measurement block.

n	M	SD	M	SD	M	SD	M	SD	M SD	M SD
Con. 23	75.54	24.23	79.53	19.57	205.26	148.90	167.57	111.26		
L-S 32	76.79	16.38	84.05	17.09	251.01	127.60	168.68	88.61		4.19 0.79
L-M 35	76.80	19.93	79.85	15.85	191.94	92.84	181.73	230.66		3.62 0.75
E-S 24	74.43	21.60	78.43	23.20	203.08	127.21	120.95	60.46	3.67 1.15	
E-M 35	74.82	19.55	83.09	20.01	192.97	117.83	128.64	107.65	4.01 0.68	

Note. $N = 149$.
Con = Control, LS = Late Assessment, Single Item, LM = Late Assessment, Multi Item, ES = Early Assessment, Single Item, EM = Early Assessment, Multi Item.

calculated the Cohen's Kappa coefficients ($M = 62.43$, $SD = 39.45$). Afterwards, we calculated the task completion time ($M = 244.94$, $SD = 139.74$) using the time the participants spent on the task for the baseline measure. The descriptive data for all groups divided into the two blocks is depicted in Table 1.

5.2. The effect of trust assessment on dependence (H1)

To analyze the effect of trust assessment on dependence, we compared dependence between all groups in a one-factor ANOVA for each task block, separately.

Recommendation Agreement. For recommendation agreement in task block 1, we found no difference between control (Con; $M = 75.54$, $SD = 24.23$), single item (L-S & E-S; $M = 75.78$, $SD = 18.65$) and multi-item (L-M & E-M; $M = 75.81$, $SD = 19.63$) groups ($F(2, 146) = 0.002$, $p = .998$). For the model the calculated η^2 value of 0.00002 indicates no effect according to Cohen.

In parallel, for task block 2, we did not find significant difference in dependence measurement between control (Con; $M = 79.53$, $SD = 19.57$), single item (L-S & E-S; $M = 81.64$, $SD = 19.94$) and multi-item (L-M & E-M; $M = 81.47$, $SD = 17.99$) groups ($F(2, 146) = 0.11$, $p = .895$). For the model the calculated η^2 value of 0.002 indicates a very small effect.

Task completion time. For the time on task comparison between the groups in task block 1, we found no difference between the groups (Con; $M = 205.26$, $SD = 148.90$, L-S & E-S; $M = 230.47$, $SD = 128.52$, L-M & E-M; $M = 192.46$, $SD = 105.30$) as well ($F(2, 146) = 1.53$, $p = .220$). Here, the generalised η^2 value of 0.02 would show a small effect.

For the task completion time between groups (Con; $M = 167.57$, $SD = 111.26$, L-S & E-S; $M = 148.22$, $SD = 80.76$, L-M & E-M; $M = 155.19$, $SD = 180.67$), we did not find a significant difference between the groups regarding the time needed to complete the tasks in block 2 ($F(2, 146) = 0.16$, $p = .856$). Here, the generalised η^2 value of 0.002 showed a very small effect.

Based on the results of the ANOVAs for the dependence measure in both blocks, there is no sufficient evidence to support H1, stating that there is a difference in dependence after trust assessment has been carried out. The results are depicted in Figure 3.

5.3. The effect of trust assessment extent on dependence (H2)

Recommendation Agreement. We compared the dependence of all groups that were assessed with single-item (L-S & E-S; $M = 78.37$, $SD = 17.40$) and all groups that were assessed with a multi-item (L-M & E-M; $M = 80.03$, $SD = 16.18$) questionnaire. The recommendation agreement was calculated based on the responses of both task blocks. Due to the larger sample size we conducted a t-test even though the normality assumption was violated (Sullivan and D'Agostino 1992). Our results indicate that the observed difference ($t(105) = 0.52$, $p = .699$) is not statistically significant. The effect size (Cohen's $d = -0.1$) indicates a small effect according to (Cohen 1992). Accordingly, our results did not support the assumption of any difference in recommendation agreement between multi-item and single-item assessments. The results are depicted in Figure 4.

Task completion time. We did not find a significant difference in task completion time between single- (L-S & E-S; $M = 246.92$, $SD = 122.32$) and multi-item (L-M & E-M; $M = 226.41$, $SD = 123.74$) groups ($t(108) = -0.88$, $p = .810$). Here, Cohen's $d = 0.17$ also indicates

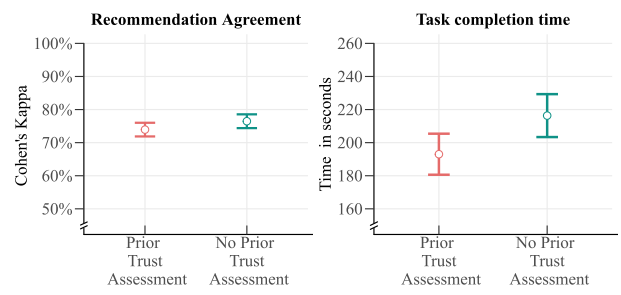

Figure 3. Comparison of user behaviour means and standard errors for the early and late trust measurement groups of the first measurement block. Task dependence is shown through the average Cohen's kappa coefficients. The time needed to complete the tasks is measured in seconds.

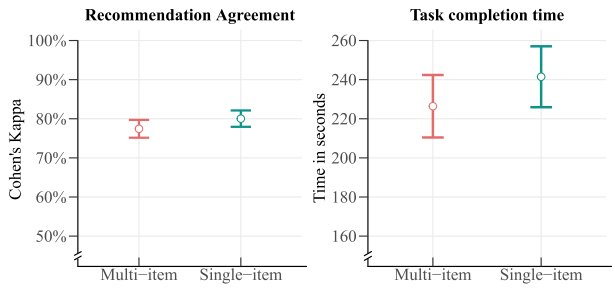

Figure 4. Comparison of user dependence measures with corresponding standard errors for multi- and single-item trust assessment groups for blocks 1 and 2 through the average Cohen's kappa coefficients for task dependence as well as the time needed to complete the tasks in seconds.

a small effect. Accordingly, based on our results H2 that the 'multi-item measure' has significantly higher task completion time than the 'single-item measure' cannot be supported.

5.4. The effect of timing of trust assessment on dependence (H3)

To assess whether the time of trust assessment affects user behaviour we differentiated between all groups that were assessed in Block 1 and all groups that were assessed in Block 2. The dependence for the blocks following immediately after the trust assessment were compared.

Recommendation agreement. To analyze the effect of time on recommendation agreement, we compared the Cohen´s kappa values of singleand multi-item measures of the first block (E-S & E-M; $M = 74.66$; $SD = 20.23$) with the Cohen´s kappa values of single- and multi-item measures of the second block (L-S & L-M; $M = 81.85$; $SD = 20.23$). The results indicate significantly lower dependence in the early assessment block than in the late assessment block ($t(112) = -2.17$, $p = .032$, Cohens' $d = -0.39$) and do therefore support H3.

Task completion time. We then compared the time needed by the participants of single- and multi-item measure groups of the first block (E-S & E-M; $M = 197.08$; $SD = 120.75$) with the participants time of the singleand multi-item groups of the second block (L-S & L-M; $M = 175.50$; $SD = 176.47$). For this comparison, the results show, that there is no significant difference between the task completion time of the blocks ($t(117) = 0.81$, $p = .420$, Cohens' $d = 0.14$), and do not support H3.

5.5. Exploratory analysis

5.5.1. Dependence development over time

To understand whether effects in dependence were based on experience over time, we additionally compared the dependence measures between the first ($M = 75.76$, $SD = 19.90$) and second ($M = 81.23$, $SD = 18.87$) block. Hence, Cohen's Kappa coefficient for both blocks was calculated and used for analysis. The development over all conditions is depicted in Figure 5 (and for Time on Task in Figure 6, respectively). We found a significant difference in dependence between the first and the second block ($t(148) = -4.77$, $p < .001$). Here, the t-test shows a small effect size of Cohen's $d = 0.28$ regarding the comparison of dependence between the first and second block. An increase can also be observed in Task 5. Since the tasks were randomised, this cannot be attributed to the content of the tasks. It is possible that the repeated use of a questionnaire may have had an effect, at least in the short term, although this effect already decreased again in the next task. A more extensive investigation with more blocks than in this experiment would be necessary to determine the effect in Task 5 (Table 2).

5.5.2. Relationship between dependence and perceived reliability

In order to understand possible other factors for dependence demonstrated by participants, we also estimated, to what extent the perceived reliability ($M = 73.09$, $SD = 20.01$) of the AI system was connected to recommendation agreement ($M = 78.16$, $SD = 18.14$) as an indicator of dependence. The Pearson correlation coefficient for the estimated AI performance and dependence of the participants shows a moderate positive correlation ($r = 0.37$, $p < .001$).

5.5.3. Prediction of dependence based on perceived reliability and trust assessment

Multiple linear regression was used to test whether perceived reliability assessed through the estimated AI performance (E-S, E-M, L-S, L-M; $M = 73.63$, $SD = 19.46$) and both single- and multi-item trust measurements (ES, E-M, L-S, L-M; $M = 3.89$, $SD = 0.86$) with a fixed intercept given by the instructed reliability of 0.8 significantly predict recommendation agreement determined through the Cohen's kappa coefficient (E-S, E-M, L-S, L-M; $M = 78.49$, $SD = 18.60$). The results show that the overall regression was not significant ($R^2 = 0.00808$, $F(2, 110) = 0.45$, $p = .640$). We can therefore conclude that predicting the recommendation agreement cannot be significantly improved by integrating perceived reliability and trust assessment.

5.5.4. Prediction of dependence based on individual differences

Another multiple linear regression model was calculated to test whether the propensity to trust (E-S, E-M, L-S, L-

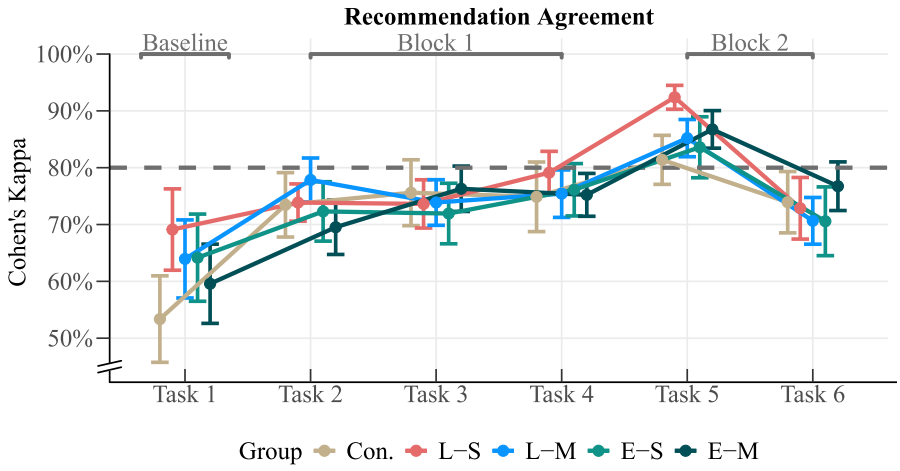

Figure 5. Comparison of recommendation agreement represented by the average Cohen's kappa coefficients and the standard error of the mean for each measurement block and task presented in the chronological sequence in which the tasks were given. Additionally, the instructed reliability of the AI system of 80% is marked.

M; $M = 3.68$, $SD = 0.69$) and affinity for technology interaction (E-S, E-M, L-S, L-M; $M = 3.32$, $SD = 1.04$) predict recommendation agreement assessed through Cohen's kappa coefficients (E-S, E-M, L-S, L-M; $M = 78.49$, $SD = 18.60$). The intercept was again given by the instructed reliability and set to the constant value of 0.8. The results of the model indicate again that the overall regression was not significant ($R^2 = 0.00405$, $F(2, 123) = 0.25$, $p = .779$).

5.5.5. Direct comparison of trust assessment

Measurement extent. To explore potential effects of assessment extent on trust values, the trust assessment of the single- (L-S & E-S; $M = 3.98$, $SD = 0.98$) and multi-item (L-M & E-M; $M = 3.82$, $SD = 0.73$) groups of both blocks were compared using the Welch two sample t-Test. The results showed, that there is no significant difference between the trust values of single- and multi-item assessment groups ($t(94) = 0.99$, $p = .324$, Cohens' $d = 0.19$).

Measurement timing. Furthermore, we compared the trust measures of the early (E-S & E-M; $M = 3.87$, $SD = 0.91$) and late (L-S & L-M; $M = 3.91$, $SD = 0.82$) measurement groups of the first block, to analyze whether the measurement time point has an impact on trust values. Again the results indicate, that the observed difference between the trust values of early and late assessment is not significant. ($t(101) = -0.26$, $p = .793$, Cohens' $d = -0.05$).

5.5.6. Correlation coefficients between dependence and trust assessments

Furthermore, the correlation coefficients between dependence and trust values were calculated for the

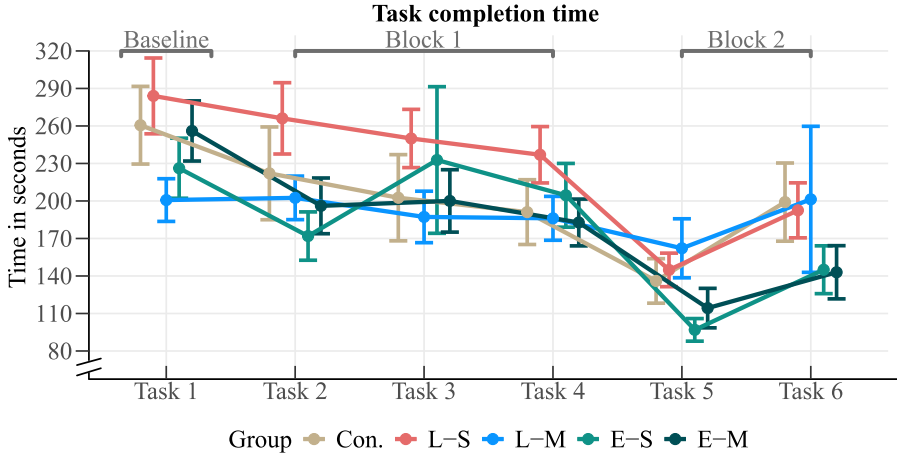

Figure 6. Comparison of task completion time for each measurement block and task presented in the chronological sequence in which the tasks were given.

Table 2. Akaike information criterion and Bayesian information criterion comparison of regression models for the prediction of recommendation agreement.

Model	*R*2	AIC	BIC
Instructed reliability	–	−64.44	−61.61
Perceived reliability	0.00002	−62.45	−56.77
Trust	0.00122	−67.29	−61.85
Propensity to trust	0.00190	−63.75	−58.09
ATI	0.00080	−63.61	−57.95

general comparison of single- and multi-item measures. With a Pearson correlation coefficient of $r = 0.45$ and $p < .001$ for the single-item measure and a $r = 0.42$ and $p < .001$ for the multi-item measure, the analysis shows a moderate positive correlation between dependence and trust values. Additionally, the Pearson correlation coefficient between the trust assessments of the multi- and single-item scales was calculated.

5.5.7. Multi-level-model based analysis of the mere-measurement effect

We conducted an exploratory analysis to test the level of trust combined with the impact of trust assessment. Specifically, we examined whether including the assessment of trust as another predictor to the level of trust could better explain participants recommendation agreement and task completion time. We assumed that trust values of participants for whom trust was not assessed are distributed similarly to the values of participants with trust assessment. Thus, we first simulated trust data for the control condition based on the distribution of trust scores within the experimental condition. The average trust score ($M = 3.89$) and its standard deviation ($SD = 0.86$) were then used to generate normally distributed trust data for measurements without trust assessment.

For both recommendation agreement and task completion time, a null model including only the fixed intercept and two baseline models including variables for the grouping context were created to explore the random effects (i.e. the variability due to grouping contexts) structure of the data and to assess the appropriateness of a multi-level approach. The first baseline model considered only participants as the grouping context, which already accounts for a considerable amount of variance based on the intraclass correlation coefficient (ICC) as a measure of variance between groups (Recommendation Agreement: ICC = .71, Task Completion Time: ICC = .58) while the second model additionally included the measurement block as a second grouping factor (Recommendation Agreement: ICC = .74, Task Completion Time: ICC = .74). To compare the goodness of fit and determine whether adding complexity to the model improves its accuracy, we used likelihood ratio tests. The results indicate that the baseline model considering participants as a grouping context is significantly better compared to a model without any random effects for both the recommendation agreement and task completion time Recommendation Agreement: $\chi^2(3) = 102.45$, $p < .001$, Task Completion Time: $\chi^2(3) = 60.94$, $p < .001$. However, in both cases, considering participants and measurement blocks shows a further improvement in explanatory power (Recommendation Agreement: $\chi^2(4) = 15.98$, $p < .001$, Task Completion Time: $\chi^2(4) = 63.14$, $p < .001$). Thus, we can assume that the multi-level modelling approach with the more complex model can be considered appropriate in both cases.

To analyze recommendation agreement and task completion time Linear Mixed-effect Models were used to account for the nested structure of repeated measures and measurement blocks. Analyses were conducted in R using the packages lme4 (Bates et al., 2015) and lmerTest (Kuznetsova et al., 2017) to fit the multi-level regression models. Satterthwaite's approximation method (Luke, 2017) for adjusted estimates of the degrees of freedom was used for the fixed effects. To allow for model comparisons through LRT, the models were fitted using maximum likelihood.

For a simple model only considering trust, we found a significant effect of trust level on recommendation agreement ($b = 0.11$, $t(148.52) = 3.89$, $p < .001$). This suggests that when the level of trust is positive, recommendation agreement increases by approximately $b = 0.11$ units compared to a negative trust level. However, the change is relatively minor compared to the range of the scale. Additionally, adding whether the trust was assessed or not does not significantly improve the model's explanatory power any further ($\chi^2(6) = 1.29$, $p = .256$).

6. Discussion

6.1. Summary of results

The objective of the present research was to evaluate and examine the magnitude of the question-behaviour effect of trust assessment on dependence in the context of an AI-supported task. Our results showed no support for the presence of the question-behaviour effect in trust assessment and dependence. That is, assessing trust in participants did not affect their subsequent dependence behaviour (H1 was not supported). Further, we did not find support for an effect of either trust assessment with single or multiple items in comparison to a control condition (H2 was not supported). While we found higher

dependence after a late trust assessment (H3 was supported partially for agreement, not for task completion time), that was potentially due to an overall increase of dependence over time. Interestingly, the predictive value of self-reported trust assessment for dependence was surprisingly low, i.e. the prediction of agreement with AI recommendations did not benefit from analyzing trust. Further exploratory analyses revealed that participants followed AI recommendations in around 80% of all decisions during the course of the study. That mirrors the instructed reliability of the AI system, as participants were told in the introduction that the AI was only correct 80% of the time and were interacting with such a system during familiarisation, but were interacting with a system with 100% reliability in the experimental task blocks. That is, participants' behaviour seems to reflect this instructed reliability. In addition, we found higher dependence in the second experimental block, compared to the first experimental block. Interestingly, single- and multi-item assessments did not differ significantly nor were they significantly correlated.

6.2. Implications

Contrary to our hypotheses, trust assessment showed no significant influence on dependence in our study. This finding was independent of scale extent, i.e. we did not find any effect of scale length on dependence, and suggests that it is possible to assess trust without affecting the users' systemrelated behaviour. Accordingly, we did not find support for any effects connected to the question-behaviour effect (Morwitz and Fitzsimons 2004; Sarah Wilding, Conner, and Sheeran 2016). However, an alternative explanation for this could be based on the comparable low correlation between trust assessment and dependence. If behaviour does not correlate as strongly with assessed attitude as initially assumed, it stands to reason that the question-behaviour effect would not be observable in this context. A low relevance of the question-behaviour effect for behaviour in contrast to higher importance for intention (Chapman 2001) has been discussed before and could also explain the results of the present study.

Due to the descriptive match between instructed reliability (80%) and dependence (mean over both task blocks and groups is 78%), instructed reliability appears to be more influential in determining behaviour compared to assessed trust. By assuming the instructed reliability as a fixed intercept in a regression analysis, adding other variables (such as trust, perceived reliability, ATI, or propensity to trust) did not yield any substantial improvement in predicting recommendation agreement. While this insight aligns in general with the broader research that emphasises the importance of perceived reliability as a key factor influencing user behaviour (e.g. Hutchinson et al. 2022), it highlights potential limitations of self-reported assessments. Consistent with the concept of 'probability matching' as described in (Bartlett and McCarley 2017), our study found that participants complied with system recommendations depending on the instructed reliability of the system across various conditions. This suggests a general tendency of users to match their dependence to the anticipated system reliability (Tikhomirov et al. 2023). This strategy can account for results in previous studies, however, there is no research on probability matching as described in the present study. While probability matching is an ineffective strategy for the given experiment (as the AI was 100% correct in the task blocks), participants potentially assumed that the instruction of 80% reliability definitely applied to the tasks presented to them. Accordingly, they may not question the instructed reliability but incorrectly assume it to be true. That is, after a series of correct answers from the AI system, they may have believed that due to the instructed reliability, an incorrect recommendation was guaranteed and looked for more closely. That behaviour, which is similar to the gambler's fallacy (Lawrence and Makridakis 1989) could explain why participants reject correct AI recommendations. Our results would support a comparable strategy in participants, which would in turn be aligned with research on probability matching (Bartlett and McCarley 2017). Further research needs to examine to what extent and under which conditions users reflect on the instructed reliability, as instructing reliability of systems might be central in shaping dependence.

Yet, in the context of our research question, the occurrence of probability matching is particularly interesting when considering the heterogeneity of trust assessment methods. These include the extent to which system performance and reliability are addressed within different assessments (e.g. in Jian, Bisantz, and Drury 2000; Korber, Baseler, and Bengler 2018 Trommler, Attig, and Franke 2018;). Therefore, the extent to which the participants' perception of reliability contributes to trust assessment could also be a factor influencing the (diverging) predictive value of the trust assessments. In our explanatory analysis, we did not find a significant correlation between dependence and the single item 'How much do you trust the system?' of the multi-item trust scale. However, the relationship between perceived reliability reported by users in percent at the end of the study and dependence within the study did not exceed the relationship between

trust assessment and dependence. This leaves the question, of which self-reported assessment – of trust, reliability, or even another related construct – may demonstrate higher predictive value for dependence. Interestingly, while the AI had 100% reliability, the majority of participants did not report perceived reliability over 80% (only 48 of 149) but mostly a perceived reliability under 80%. The massive gap between perceived and actual reliability has been already reported in earlier research (Rieger, Roesler, and Manzey 2022). Although we expected to observe a description-experience gap (see Chen et al. 2018), participants reported perceiving 80% reliability of the system, despite this perception being incorrect. While we found a higher recommendation agreement over time, there is no clear trend or indication that participants approximated higher levels of recommendation agreement based on their individual experience. That strengthens the need for further research to understand the role of instructed (or outside of research settings, reported) reliability in comparison to participant experience and perception. It needs to be noted, however, that the perceived reliability in percent was only surveyed at the end of the study and not between experimental blocks and the study was potentially too short to clearly identify effects over time (Holliday, Wilson, and Stumpf 2016). In addition, participants recruited via Profilic potentially perceived this question as the attention check and reported the instructed reliability here to keep eligible for compensation.

Our results imply that the instructed reliability of a system might be a better predictor of human dependence than self-reported trust assessment at least in tasks where AI systems give a clear recommendation for a specific action and AI recommendation is presented before participants make their own decision. Therefore, the predictive value of instructed reliability is important to consider when designing AI-based support systems: Further research needs to examine how humans estimate the correctness of information provided by systems on their reliability and factors influencing reliability when assessing dependence (e.g. by using explanations, see Nourani et al. 2019). In our experiment, no detailed information on factors for the system‘s reliability was given to the participants, for example, low confidence in the system when colour-based rules are involved. Participants' estimation of AI reliability was lower than the actual AI reliability, which has been illustrated in earlier research (Rieger et al. 2023; Rieger, Roesler, and Manzey 2022). Further research on AI-supported pattern detection should explore methods to improve users' perception of system reliability.

As demonstrated by existing studies, the lack of predictive value of trust assessment for behaviour can be contingent on the specific task design. In the present study, the AI system provided clear recommendations without considering the individual's state or hypotheses, setting it apart from evaluative AI as defined in (T. Miller 2023). As discussed before (van Dongen and van Maanen 2013), trust might be more influential on dependence when users need to act before the systems or need to utilise system information to make a decision instead of receiving a recommendation. This suggests that the role of trust and its influence on the question-behaviour effect might vary with different HAII modalities and sequences within a task. The differences in the predictive value of trust in the early group vs. the late group indicate that the role of trust could also change with growing familiarity with the system (see also Korber, Baseler, and Bengler 2018). It is possible that a different interaction dynamic between humans and AI could result in an altered role of trust and consequently, also, the question-behaviour effect. This possibility underlines the need for further research into varying types of HAII and their implications for trust dynamics as well as for the question-behaviour effect.

The results of our study show that trust research in HAII needs to examine the predictive value of established constructs when trying to reliably predict behaviour. The use of trust assessment in an interaction where the AI provides a clear recommendation, however, does not influence the subsequent dependence demonstrated and does not result in a question-behaviour effect. However, the predictive value of trust assessment remains a topic for further study (Bolton 2022), and the findings need to be applied to different types of interactions (i.e. different types of collaboration/ combination of information processing and action regulation by the AI versus the user) to verify their generalisability. In particular, it remains an open question, under which conditions the instructed reliability of a system is more important than users' experience, for example, dependent on the frequency of feedback, the sequence of AI and human actions in a shared task, or the length of an experiment. As more regulation demands AI development to provide information on an AI system's reliability and limitations (e.g. the EU AI Act, European Commission 2021) the impact on dependence of such information needs to be studied in detail.

6.3. Limitations and further research

In interpreting the results of the present research, several aspects have to be kept in mind, especially regarding the design of the presented task. The potential

complexity of Kandinsky Tasks may have been challenging for participants, limiting the interpretability of recommendation agreement and affecting the comparability of results with other studies, that utilise easier tasks. As participants' baseline (i.e. the number of tasks solved without the AI system) was at 60% correctly solved Kandinsky Tasks on average, the goal of 90% may have been an unrealistic high demand to function as an additional incentive. If participants did not expect to be able to meet the criterion for additional reward even with AI help, the vulnerability necessary to experience trust (Lee and See 2004) was potentially not present. As previous research demonstrated, trust might decline when automated systems are not reliable enough (Moray, Inagaki, and Itoh 2000). For the attitude of the participants to be sufficiently strong to influence behaviour, a better performing system (i.e. with reliability levels of 90% and more) could be needed. In addition, previous research demonstrated that only medium or low levels of an attitude may have drastically fewer effects than strong attitudes (see Kim, Chun, and Song 2009). Differentiating to what extent participants were 'giving up' and just following the AI recommendation is difficult to examine. However, as the difficulty was the same in all conditions, we do not believe that this impacted our experimental manipulation to a great extent. However, our exploratory results remain limited to tasks with higher complexity and strict AI recommendations (i.e. the AI system does not deliver information to determine the correct answer, but recommends specific answers directly).

It is important to note the specificity of the task as well as its abstract nature. While it was essential for our research to have a task where users could not differ in pre-existing knowledge, collaborative human-AI tasks involve experienced humans, and expertise can change the result of the interaction (Inkpen et al. 2023). Previous studies already demonstrated how assigning similar tasks to human and AI systems may lead to decreased performance of human users, comparable to social loafing (Cymek, Truckenbrodt, and Onnasch 2023; Inuwa-Dutse et al. 2023). Accordingly, future research should aim to present tasks that require, for example, a combination of skills or exchange of information (see Bard et al. 2020).

As another important point, further research is needed to investigate varied levels of automation and interaction types to understand how these factors impact trust assessment and dependence. For example, users always saw the AI recommendation at the same time they saw the Kandinsky Figures – changing the sequence of decision-making could change the role of trust or even improve decision-making in general (see Dzindolet et al. 2002 compared to Rieger et al. 2023).

7. Conclusion

In conclusion, the present study contributes to the understanding of trust dynamics in human-AI interaction and raises further questions for trust assessment and communication of information on the reliability of AI systems. Contrary to our initial hypotheses, we found that self-reported trust assessment has no discernible impact on reliance behaviour in AI-supported tasks. This observation is underpinned by the absence of substantial predictive power of trust assessment in forecasting reliance. Notably, the strongest predictor of behaviour in our experimental setup was the instructed reliability of the AI system, which is more relevant than self-reported trust assessment or perceived reliability. These findings suggest a need for a reevaluation of the role of trust assessment in AI interactions and point toward the greater influence of perceived system reliability on user behaviour.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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